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📍 Dhaka, Bangladesh

Experience

Software Development Engineer 2

May 2026 – Present

IQVIA — AI & Technology Solutions

Dhaka, Bangladesh

- Promoted from SDE 1 in recognition of consistently exceeding expectations across AI delivery, cross-team collaboration, and technical ownership throughout a critical product expansion phase.
- Leading architectural extension of the AI-powered Data Explorer and Collaboration Board ecosystem — expanding scope into agentic system design, MCP server lifecycle management, and backend platform scalability.

Software Development Engineer 1

Jul 2024 – Apr 2026

IQVIA — AI & Technology Solutions

Dhaka, Bangladesh

- Architected the integration of **Data Explorer** and **Slide Builder** into IQVIA's AI Assistant via **Model Context Protocol (MCP)** — built MCP servers for Collaboration Board, Export Tools, and Data Explorer Agent, enabling natural language-driven chart generation, slide creation, and report export.
- Configured agentic tools (chart/table edits, widget search, PPT template application, GraphQL execution) into **Dual Node** and **Orchestration Agent** workflows for end-to-end AI-driven enterprise presentation generation.
- Optimized AI-to-Collaboration Board workflow by ~50% and slide deck population performance by ~60% through targeted backend refactoring and agent server restructuring.
- Built the **AI-Generated Textual Insights Refresh** pipeline from POC to production — agent call flows, streaming APIs, multi-tenant deployment, and prompt engineering for Orchestration and Dual Node agents.
- Delivered **full AI streaming support** inside Slide Builder (OmniboxCall, SummarizeSlide, SummarizeSlideDeck APIs); extended GraphQL APIs for data source configuration, AI Assistant integration, and user/team navigation.
- Built end-to-end **User Report Subscription** system, **Default Collaboration Board Agent**, Dashboard-to-Slides export, and live text auto-update among several other major roadmap features.
- Rated “*Significantly Exceeds Expectations*” — manager described delivery as “*absolutely phenomenal*” under high autonomy, undocumented technology, and unclear requirements.

Publication

5GPT: 5G Vulnerability Detection via Zero-Shot GPT-4 & Domain-Aware Prompt Engineering

Jul 2025

IEEE Transactions on Information Forensics & Security — Q1, Impact Factor: 8.0

- A. Shahriar, [S.J. Hisham](#), K.M.A. Rahman, R. Islam, M.S. Hossain, R.H. Hwang, Y.D. Lin  ieeexplore.ieee.org/document/11072222

Selected Projects

Agentarium & Evolverium | *Python, LangGraph, MCP, CrewAI, AutoGen, OpenAI Agents SDK, Gradio* 

- 14+ autonomous AI agents (trading, coding, research, storytelling) and 7 advanced agentic systems — genetic loops, adversarial debate, Bayesian cascades, plan-then-execute — with custom MCP servers, persistent memory, and Gradio UI.

DRL Renegades & Advanced DRL Renegades | *Python, PyTorch, Stable-Baselines3, gym, Unity ML-Agents* 

- Deep RL agents for Street Fighter III (PPO/A2C), Mortal Kombat, Rubik's Cube (DQN), Pacman, Super Mario Bros, and classic Atari environments. Advanced version extends with curriculum learning, reward shaping, self-play, and multi-agent training pipelines.

Robo-Chess | *Python, PyTorch, Tianshou, Stockfish, gym-chess* 

- Chess AI combining AlphaZero+MCTS for full-game strategy, DQN/DDQN for endgame tablebase solving, and self-play imitation learning from Lichess grandmaster games. Best model achieves ~600 ELO Stockfish level with a dedicated Kaggle competition entry.

LinkGenius-AI | *React 19, TypeScript, Express.js, SQLite, Gemini Flash, Tailwind CSS* 

- AI-powered URL shortener featuring an AI Assistant, real-time site preview, bulk URL shortening, QR code generation, detailed click analytics, IP-based rate limiting, and malicious URL detection via the Google Safe Browsing API.

HallSync | *React, Next.js, TypeScript, Node.js, PostgreSQL, Prisma ORM, Apollo GraphQL* 

- Full-stack university hall management system with automated seat allocation, smart meal management, complaint tracking, announcement broadcasting, and a role-based statistics engine. **Selected as one of the top 5 projects** in the academic cohort.

Subset C Compiler | *C++, Bison (Yacc), Flex (Lex), x86 Assembly* 

- Full compiler pipeline for a subset of C — implementing a Symbol Table Manager, Lexical Analyzer, Syntax & Semantic Analyzer with error recovery, and an Intermediate Code Generator producing optimized x86 Assembly output.

Education

Bangladesh University of Engineering & Technology (BUET)

Dhaka, Bangladesh

M.Sc in Computer Science & Engineering — Part-time, Ongoing | *CGPA: 3.75 / 4.00*

Sep 2024 – Present

Bangladesh University of Engineering & Technology (BUET)

Dhaka, Bangladesh

B.Sc in Computer Science & Engineering | *CGPA: 3.82 / 4.00*

Apr 2019 – Jun 2024

Technical Skills

Languages: C#, Python, Java, C/C++, TypeScript, JavaScript, SQL

Frameworks & Backend: .NET, ASP.NET, Node.js, Express.js, GraphQL, React, Next.js, PyTorch

AI / Agentic / ML: LangGraph, MCP, CrewAI, AutoGen, OpenAI Agents SDK, Scikit-Learn, NumPy, Pandas

Databases: PostgreSQL, MongoDB, Oracle, SQLite, Prisma ORM